

22 A battery causes a current of 3 A to flow through a metal wire of diameter 2.0 mm. The number density of the free electrons in the metal wire is $8.5 \times 10^{28} \text{ m}^{-3}$.

What is the average speed of the electrons drifting along the wire?

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| A | $2.0 \times 10^{-11} \text{ m s}^{-1}$ | B | $2.0 \times 10^{-5} \text{ m s}^{-1}$ | C | $7.0 \times 10^{-11} \text{ m s}^{-1}$ | D | $7.0 \times 10^{-5} \text{ m s}^{-1}$ |
|----------|--|----------|---------------------------------------|----------|--|----------|---------------------------------------|

Answer: D

$$I = nAvq$$

$$3 = (8.5 \times 10^{28})(\pi \times 0.0010^2)v(1.60 \times 10^{-19})$$

$$v = 7.0 \times 10^{-5} \text{ m s}^{-1}$$

23 A heater of resistance 5.0 Ω dissipates 125 W of power when current flows through it. The